

VirusTotal

# How to Use VirusTotal for Sandbox Analysis

A Visual Step-by-Step Guide  
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Open VT

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Upload File

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Detection

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Live Demo

Analyse suspicious files, domains, IPs and URLs

FILE

URL

SEARCH

☁ Drag and drop a file here



Choose file

Max file size: 650 MB

3 analysis modes:  
FILE · URL · SEARCH

Click here to choose  
your file — or drag & drop

**400M+**

Files analysed

**200M+**

Unique URLs

**70+**

AV engines

FILE

URL

SEARCH

suspicious\_file.exe

×

File selected — ready to upload

↓

Confirm upload

Analysing... 43%

Running 72 AV engines simultaneously · Detonating in sandbox · ~45 seconds remaining

Kaspersky

BitDefender

Symantec

McAfee

ESET

Avast

- File chip appears — confirm it is correct
- Click this to start! Malware runs in their cloud, NOT on your machine.
- Progress bar — 72 engines scanning simultaneously

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY



/ 72 security vendors and 1 sandbox  
flagged this file as malicious

47 out of 72 engines  
flagged this as malicious!  
Red = BAD

suspicious\_file.exe · SHA-256: a1b2c3d4e5f6... · 452 KB

| ENGINE        | CATEGORY   | RESULT                       |
|---------------|------------|------------------------------|
| ! Kaspersky   | malware    | Trojan.Win32.GenericML       |
| ! BitDefender | malware    | Gen:Variant.Mikey.117248     |
| ! ESET-NOD32  | malware    | Win32/TrojanDownloader.Agent |
| ! Avast       | malware    | Win32:TrojanX-gen [Trj]      |
| ✓ Symantec    | undetected | (no detection)               |
| ! Microsoft   | malware    | Trojan:Win32/Wacatac.B!ml    |
| ✓ McAfee      | undetected | (no detection)               |

Each row = 1 AV engine  
Red = detected threat  
Green = no detection

Even 25 engines  
missed it — zero-day  
gap in signatures!

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY

Sandbox:

VirusTotal Jujubox

CAPE Sandbox

Zenbox

Network Communications

8 connections

api.unknown-domain.ru · 185.44.81.22:4444 · POST /gate.php

Files Written

12 files

C:\Users\user\AppData\Roaming\svchost32.exe

Registry Keys Set

3 keys

HKCU\...\CurrentVersion\Run → 'WindowsUpdate'

Processes Spawned

5 processes

cmd.exe → powershell.exe -enc [base64] → mshta.exe

Signature Matches

7 matched

Execution via PowerShell · Process Injection · C2 Beacons

MITRE ATT&CK

6 techniques

T1059.001 · T1547.001 · T1071.001 · T1082

Click BEHAVIOR tab — this is the sandbox execution data!

RED = very suspicious! 8 outbound connections to unknown servers

Persistence! Malware added itself to auto-run on startup via registry

MITRE ATT&CK tells us EXACTLY what attack techniques were used

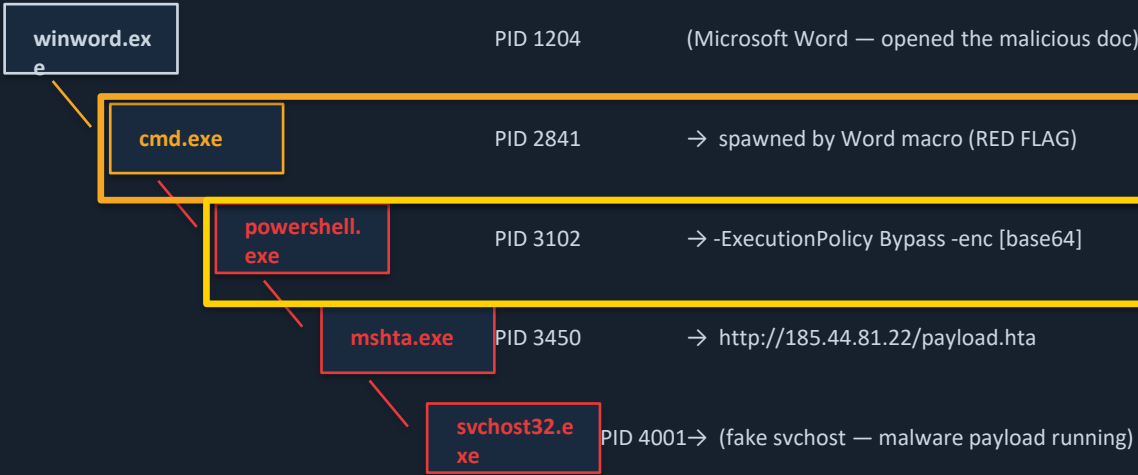
| Network Communications (8 connections detected) |                          |  |      |         |
|-------------------------------------------------|--------------------------|--|------|---------|
| TYPE                                            | DESTINATION              |  | PORT | DETAILS |
| DNS                                             | api.unknown-ru.com       |  | 53   |         |
| Domain lookup — C2 server address               |                          |  |      |         |
| HTTP                                            | 185.44.81.22             |  | 80   |         |
| GET /payload_stage2.bin — DOWNLOAD              |                          |  |      |         |
| HTTPS                                           | 185.44.81.22             |  | 443  |         |
| POST /gate.php — SENDING DATA OUT               |                          |  |      |         |
| TCP                                             | 91.108.56.199            |  | 4444 |         |
| Raw socket — Metasploit default port!           |                          |  |      |         |
| DNS                                             | update.windows-patch.net |  | 53   |         |
| Typosquat — fake Windows update domain          |                          |  |      |         |
| HTTP                                            | 93.184.216.34            |  | 80   |         |
| GET /icon.png — benign (false positive)         |                          |  |      |         |

POST to /gate.php = data being SENT OUT — possible exfiltration

Port 4444 = Metasploit default! Attacker has active control channel

Typosquatting domain! 'windows-patch.net' ≠ microsoft.com — FAKE

Process Tree (5 processes spawned)



Word spawned cmd.exe!  
Normal Word docs NEVER  
do this — macro attack

PowerShell -enc = encoded  
command to hide what  
it is executing

Registry Keys Set (3 keys written)

|                                                    |                                                      |
|----------------------------------------------------|------------------------------------------------------|
| HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run |                                                      |
| Name: WindowsUpdate                                | Value: C:\Users\user\AppData\Roaming\svchost32.exe   |
| HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run |                                                      |
| Name: OneDriveSync                                 | Value: C:\Users\user\AppData\Roaming\sync_helper.exe |
| HKCU\SOFTWARE\7-Zip\FM                             |                                                      |
| Name: ShowDots                                     | Value: 1                                             |

Auto-run persistence!  
Every time user logs in  
malware restarts itself

| Contacted Domains (3)                                          |                            |                   |
|----------------------------------------------------------------|----------------------------|-------------------|
| api.unknown-ru.com                                             |                            | Malicious · 47/72 |
| update.windows-patch.net                                       |                            | Malicious · 23/72 |
| 93.184.216.34                                                  |                            | Undetected · 0/72 |
| Files Dropped (2)                                              |                            |                   |
| svchost32.exe                                                  | 452 KB · SHA256: 9f3a2b... | 47/72 Malicious   |
| sync_helper.exe                                                | 128 KB · SHA256: c4d8e1... | 31/72 Malicious   |
| ITW (In-The-Wild) URLs — where this file was hosted online (1) |                            |                   |
| http://185.44.81.22/downloads/update.exe                       |                            | 47/72             |

Click any domain to open ITS analysis — chain investigation!

Dropped files have their own VT links — click to analyse the payload too

This URL is WHERE the malware was being distributed from



 MITRE ATT&CK Techniques Detected (6 techniques)

T1059.001

Execution

Command and Scripting Interpreter: PowerShell

Malware ran a base64-encoded PowerShell command to hide its actions

T1547.001

Persistence

Boot/Logon Autostart: Registry Run Keys

Added svchost32.exe to HKCU\...\Run for persistence across reboots

T1071.001

Command & Control

Application Layer Protocol: Web Protocols

Used HTTP/HTTPS to communicate with C2 server at 185.44.81.22

T1082

Discovery

System Information Discovery

Queried OS version, hostname, user info — standard sandbox evasion check

T1055

Defence Evasion

Process Injection

Injected code into legitimate svchost.exe to hide malicious activity

T1041

Exfiltration

Exfiltration Over C2 Channel

Sent collected data back via the existing HTTP C2 connection (POST /gate.php)

 T1059 = PowerShell abuse — most common malware technique in 2024

MITRE ATT&CK = global standard — every security tool uses these IDs

 Click any technique ID to read the full MITRE description + defences

# Without Sandbox vs. With Sandbox

## ✗ WITHOUT Sandbox

- ✗ Antivirus checks file signature only
- ✗ Zero-day malware = UNDETECTED
- ✗ File runs on your real machine
- ✗ You see: 0/72 detected → assume safe
- ✗ Malware installs, phones home, persists
- ✗ You find out when it's too late
- ✗ No visibility into what it actually did
- ✗ No network IOCs to block

VS

## ✓ WITH Sandbox (VirusTotal)

- ✓ File is EXECUTED in isolated cloud VM
- ✓ Behavioral analysis catches zero-days
- ✓ Nothing runs on your real machine — ever
- ✓ Full network activity log captured
- ✓ Files dropped, registry changes — all logged
- ✓ MITRE ATT&CK techniques identified
- ✓ Exact IOCs extracted for firewall blocking
- ✓ Removal instructions from registry/process data